

salt 'myminion' state.highstate (Flow: top -> nginx -> top)

The summary output looks like what's shown below:

```
Summary
-----
Succeeded: 1
Failed:    0
-----
Total:    1
```



Note: Instead of *nginx.sls*, we can also use *nginx/init.sls*, if we have other modules common to the Nginx module and if we want to call all or some of those modules within *init*.

SLS namespace rules

Salt has defined a few simple rules for *.sls* files.

1. The *.sls* is discarded (i.e., *nginx.sls* becomes *nginx*).
2. A file called *init.sls* in a sub-directory is referred to by the path of the directory. So, *nginx/init.sls* is referred to as *nginx.dev*.
3. Sub-directories can be used for better organisation.
 - a. Each sub-directory is represented by a dot.
 - b. *nginx/dev.sls* is referred to as *nginx.dev*.
4. If both *nginx.sls* and *nginx/init.sls* happen to exist, *nginx/init.sls* will be ignored, and *nginx.sls* will be preferred over *nginx/init.sls* and read as *nginx*.

To start a service, type:

```
/srv/salt/nginx/init.sls:
```

```
nginx:
  pkg:
    - installed
  service:
    - running
    - require:
      - pkg: nginx
```

Now we have added a little more information about our preferences.

- A check is done whether a package named Nginx is installed or not. If not, it will be installed.
 - If the package is installed, then a check is done to find out whether the Nginx service is running. If not, the service is started. For the service check to be done, the package installation is marked as a dependency. The word 'require', being a Requisite Statement, does just that—ensuring that the pre-requisites are there.
- To add users and groups, type:

```
nginx:
  pkg:
    - installed
```

```
service:
  - running
  - enable: True
  - watch:
    - pkg: nginx
    - file: /etc/nginx/nginx.conf
    - user: nginx
user.present:
  - uid: 151
  - gid: 151
  - home: /var/lib/nginx
  - shell: /bin/nologin
  - require:
    - group: nginx
group.present:
  - gid: 151
  - require:
    - pkg: nginx
```

'Watch' and 'require'

We hope you noticed that we have used the 'watch' word this time. The 'require' word will make sure the user gets created after the group creation, which must happen only after the Nginx package is installed. The 'watch' word checks for any changes to the watched states. The service state watcher will be activated in the following cases:

- When the package is updated
- When the configuration file is changed
- If the user ID gets modified

The service state watcher just restarts the service; in this case, a change in the *config* file will also trigger a restart of the respective service.

Users and groups

User Nginx is created with:

- userid and groupid 151
- Home folder */var/lib/nginx*
- No login shell
- Requires a group called *nginx*

Group Nginx is dependent on the package *nginx* being installed.

Repository

The Salt master could act as the repository for the Nginx configuration file. The '*file.managed*' module sends files to the minion and replaces the configuration file on the target system.

```
/etc/nginx/nginx.conf:
file.managed:
  - source: salt://nginx/files/nginx.conf
  - user: root
  - group: root
  - mode: 644
```

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